

ONC chain: conflict with an oncoming vehicle

Derivation runs downward, from the collision (lagging SPI) back to the earliest observable signals

LAGGING SPI 7

(No. of times (ONC_COLL) per 100k manoeuvres) < T_ONC

Metric: collisions with an oncoming vehicle, per 100k manoeuvres. Threshold: T_ONC.

why

PROXIMATE CAUSE

The ego occupied the conflict area while an oncoming vehicle was closing on it, and no avoidance action succeeded in time.

reaction tier (last-resort, nearest the loss)

INTERMEDIATE LEADING SPI 5

(No. of times (AEB_ONC) per 100k) < T_AEB_ONC

Metric: manoeuvres with a hard emergency brake while an oncoming vehicle was within the collision horizon.

Mitigation attempted, collision not yet realised.

decompose the proximate cause (kinematic precursor tier)

OR GATE — either cause alone is sufficient

CAUSE A

The accepted gap was already insufficient when the manoeuvre began.
Planning subsystem.

CAUSE B

Closing speed exceeded what the turn arc allowed because of increased oncoming velocity, so the deceleration needed rose beyond normal braking.
Prediction subsystem.

SIGNAL A

TTC to the nearest closing oncoming vehicle, sampled each frame.

SIGNAL B

Deceleration required to stop short of the conflict area at the current speed.

observability and independence check

LEADING SPI 1

(No. of times (Gap_TTC < θ_{GAP}) per 100k) < T_GAP

Metric: minimum gap TTC over the manoeuvre.

Event θ_{GAP} (lower tail); Rate T_GAP.

LEADING SPI 2

(No. of times (Req_Decel > θ_{ReqD}) per 100k) < T_ReqD

Metric: maximum required deceleration over the manoeuvre.

Event θ_{ReqD} (upper tail); Rate T_ReqD.

ATTACH TO THE SAFETY CASE

SPI 1 and SPI 2 as leading evidence on the oncoming-conflict goal.

SPI 5 as intermediate leading and SPI 7 as lagging evidence on the same goal.

VALIDATE

Temporal precedence checked against SPI 7 only. Incidents are attributed by actor identity, so a pedestrian or a rear impact is never counted here. The metric is separated between safe and ONC-collision manoeuvres.

SPI-5 attaches directly, independent of the kinematic tier